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Exploring the role of ChatGPT in higher education: opportunities, challenges and ethical considerations

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Abstract

Purpose – This paper aims to examine the opportunities and challenges of using ChatGPT in higher education. Furthermore, it is also discuss the potential risks and plunders of these tools.
Design/methodology/approach – The paper discuss the use of artificial intelligence (AI) in academia and explores the opportunities and challenges of using ChatGPT in higher education. It also highlights the difficulties of detecting and preventing academic dishonesty and suggests strategies that universities can adopt to ensure ethical and useful use of these tools.
Findings – The paper concludes that while the use of AI tools, ChatGPT in higher education presents both opportunities and challenges. The universities can effectively address these concerns by taking a proactive and ethical approach to the use of these tools. This paper further suggests that universities should develop policies and procedures, provide training and support, to detect and prevent cheating intentions.
Originality/value – The paper provides insights into the opportunities and challenges of using ChatGPT in higher education, as well as strategies for addressing concerns related to academic dishonesty. The paper further adds importance to the discussion on the ethical and responsible use of AI tools in higher education.

Keywords Artificial intelligence, ChatGPT, Higher education
Paper type Research paper

Introduction

The starter of new educational technology frequently evokes strong responses, spanning from negative forecasts to uncontained enthusiasm. This pattern is also evident in the cases of both generative pre-trained transformer (GPT)-3 and ChatGPT which are built upon the GPT-3 basis. The launch of GPT-3 brought about a variety of reactions from media sources, with some conveying excitement and others voicing concerns. To exemplify, Manjoo (2020) of the British Broadcasting Corporation (BBC) questioned whether GPT-3 signaled the dawn of our AI-driven future, whereas the New York Times alerted that machines are becoming remarkably proficient at writing. According to The Economist (2020), GPT-3 at times displays a upsettingly human-like quality, for better or for inferior, while the telegraph emphasized the need for significant uneasiness regarding artificial intelligence (AI)-generated text. There is an excessive amount of excitement surrounding GPT-3.It still has

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significant flaws and occasionally produces absurd errors. Though AI will revolutionize the world, GPT-3 is just an initial preview, and we have much more to learn from their use.

In addition, there were those who were not impressed with GPT-3 (Marcus and Davis, 2020). They expressed their skepticism, stating that the program is unreliable and merely spouts meaningless statements. Kantrowitz (2022) stated that when ChatGPT first appeared, people praised it for being both petrifyingly amazing and hilariously enjoyable, as well as for successfully passing the "Nazi Test" by being "not particularly malevolent". According the genuine version of something that was previously exaggerated has arrived to (Kantrowitz, 2022). Atlantic newspaper reported that ChatGPT is part of a technological phenomenon involving generative AI that could potentially transform our perception of work, cognition and human creativity (Thompson, 2022). Metz (2022), New York Times referred to the potential to "alter the course of the world" Grant and Metz (2022), referredChatGPT as "the next major disruptor in the industry". According to an article in the newspaper, When ChatGPT first grown reputation, a lot of its admirers expressed their amazement and admiration using exaggerated language, as if they believed it to be a combination of advanced technology and magical powers (Rose, 2022). ChatGPT's potential to disrupt the dominance of Google search has impressed Alphabet's management to the extent that they have triggered a 'code red (Tung, 2023). OpenAI technology is already being used by Microsoft to enhance its own search engine, Bing.

The development of educational technology, numerous technological advancements has been envisioned as a way to replace conventional education (Rudolph, 2018; Kefalaki *et al.*, 2022). These thoughts were often driven by an enthusiastic and irrational obsession with technology. Starting from the early 1900s, different technological advancements such as movies, radio, television, computers, the Internet, mobile devices, social media and virtual and augmented reality has been recognized as game-changers in the field of education and instruction (Terzian, 2019; Tan, 2019; Kuleto *et al.*, 2021). In the earlier, when it comes to EdTech, there was often a lack of courtesy paid to how teachers utilized and how students engaged with these resources.

Different types machines have drastically altered various aspects of daily life in the 20th century, a visitor from the 19th century would still find a modern classroom quite familiar. Furthermore, this statement still holds true as physical classrooms have not undergone any significant changes in their traditional learning environment (Ferster, 2014, p. 1). For a long time, people have either careful technology to be a cure-all solution or a source of problems Rudolph (2018). But it's important to steer clear of both the belief that technology determines everything and the opposition to technology, known as Luddism. There is no amazing solution that comes from a "Magister ex machina" or a master machine. Rudolph (2014), expectations of revolutionary transformation in higher education are frequently overstated. Recent illustration of this is the massive open online courses, which were believed to signal the demise of higher education. Students still highly value credentials, which are exclusively provided by universities.

In this article, we briefly discuss the history of OpenAI, the organization behind ChatGPT. We highlight the fundamental opportunities, challenges of ChatGPT in higher education. In addition, how university faculty design assessments to prevent or minimize the use of ChatGPT by students. Lastly, we also discuss how academics can prevent students from plagiarizing using ChatGPT.

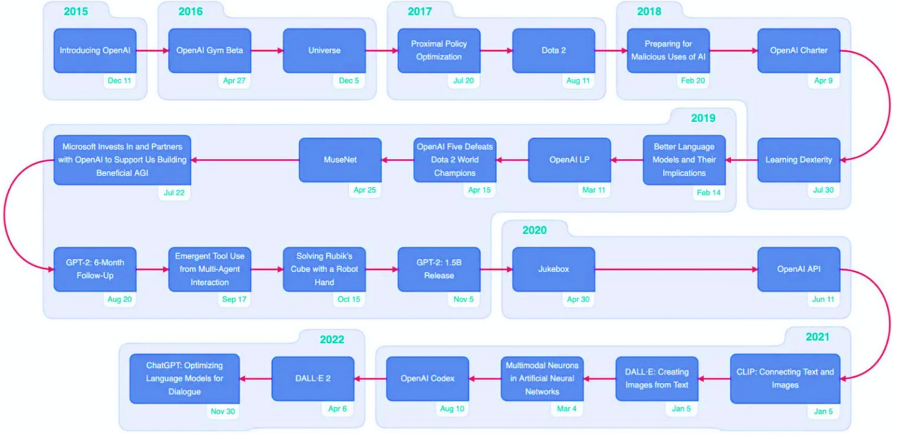
History of OpenAI and ChatGTP OpenAI

OpenAI is a research capability that focuses on the development and advancement of "friendly AI" with the aim of benefiting humanity in general (OpenAI, 2015). Their objective is to conduct research that promotes the positive bearings of artificial intelligence. OpenAI, which is headquartered in San Francisco, was established by a group of well-known tech magnates from Silicon Valley, including Elon Musk, who resigned from the organization's

Board of Directors in 2018. Reid Hoffman, the founder of LinkedIn, Peter Thiel, the co-founder of PayPal, Greg Brockman, the former Chief Technology Officer of Stripe, and Sam Altman, the founder of Y Combinator, whose business incubator aided in the growth of companies like Airbnb, Dropbox and Coinbase. Brockman and Altman presently hold the positions of OpenAI's President and Chief Executive officer (CEO). Hao's (2020) claimed that OpenAI has rapidly emerged as one of the foremost AI research laboratories worldwide, alongside entities such as DeepMind by Alphabet.

OpenAI's ultimate objective to develop an "artificial general intelligence" (AGI), which is also referred to as "strong AI" and is considered the pinnacle of AI (Searle, 1980; Grace *et al.*, 2018). AGI refers to machines that can do any intellectual task that humans are capable of performing. Markoff (2015) reported stated that the founders of OpenAI believe that AI presents a significant prospect to enhance the world, with a wide range of uses across various industries, including self-driving cars and precise tailored medicine. He further stated that the founders of OpenAI believe that AI presents a significant prospect to enhance the world, with a wide range of uses across various industries, including self-driving cars and precisely tailored medicine. Musk is worry about the risk of AI rising against humans. He has referred to it as our "most significant existential danger" and stated that we are conjuring up a demon with it, as mentioned by Markoff's (2015). Musk further emphasized that the priority should be to develop technologies that enhance human capabilities rather than replace them. As per the vision of OpenAI's creator AI was supposed to be created in a manner that is advantageous and secure for humanity (Markoff, 2015). Furthermore, they planned to publicly distribute open-source software and sophisticated AI tools without any intellectual property limitations. Hao (2020), at first, OpenAI declared that it would not be influenced by profit-based financial motives, and as a result, it would be in a good position to guide the technology with the welfare of humanity as its top priority. The proper time-line of OpenAI is presented in Figure 1 respectively. The proper credit line has been included.

OpenAI achieved a significant advancement in AI in 2020 through the introduction of the GPT-3 (Cooper, 2021). GPT-3's training involved the use of a huge amount of text, explicitly hundreds of billions of words, which equates to 45 terabytes of data. Brown *et al.* (2020), data used for this comes from various sources, including common crawl, which is a nonprofit organization that crawls the Internet and shares its archives and datasets for free with the



Source(s): OpenAI (ChatGPT) Timeline (openai.com)

Figure 1.
Timeline of OpenAI

public. In addition, Brown *et al.* (2020), there are two internet-based books corpora called Books1 and Books2, as well as information from Wikipedia. Presently, Heaven (2020), GPT-3 stands as the most significant and potent language model ever designed, utilizing deep learning to create various forms of written text, such as essays, stories, poems and code. Grossman (2020), an astonishing feature of GPT-3 is its ability to accomplish multiple tasks without specific training. In contrast, natural language processing (NLP) systems typically demand a arduous and expensive "supervised" learning method, which involves the labeling of every data piece from a vast collection of text. Grossman (2020), GPT-3 has the ability to learn from any kind of text, and without any extra training, it can accomplish a wide range of tasks. These tasks include generating stories, coding, completing images, translating languages and performing calculations. Figure 2 presented the current overview of ChatGPT statistics. The proper credit line has been included.

The GPT 3 represents a significant improvement over previous GPT models (Grossman, 2020). The size of a language model is crucial, and GPT-3 has 175bn parameters, which is much larger than the 1.5bn parameters of GPT-2 and the 110m parameters of GPT, according to several sources (Lauret, 2020; Heaven, 2020). According to Scott (2020), the training took place on the AI supercomputer of Microsoft Azure and its approximate cost is around $12m. Floridi and Chiriatti (2020), GPT-3 can be used for various aims such as translating, summarizing, answering questions, correcting grammar, creating chatbots, writing emails and many other several tasks.

According to Brown *et al.* (2020), GPT-3 has the ability to engage in zero-shot, one-shot and few-shot (FS) learning. When operating under a FS scenario, the language model is presented with a limited number of instances or demonstrations. On the other side, a situation where only one example is provided (known as a one-shot setting) is different. Additionally,

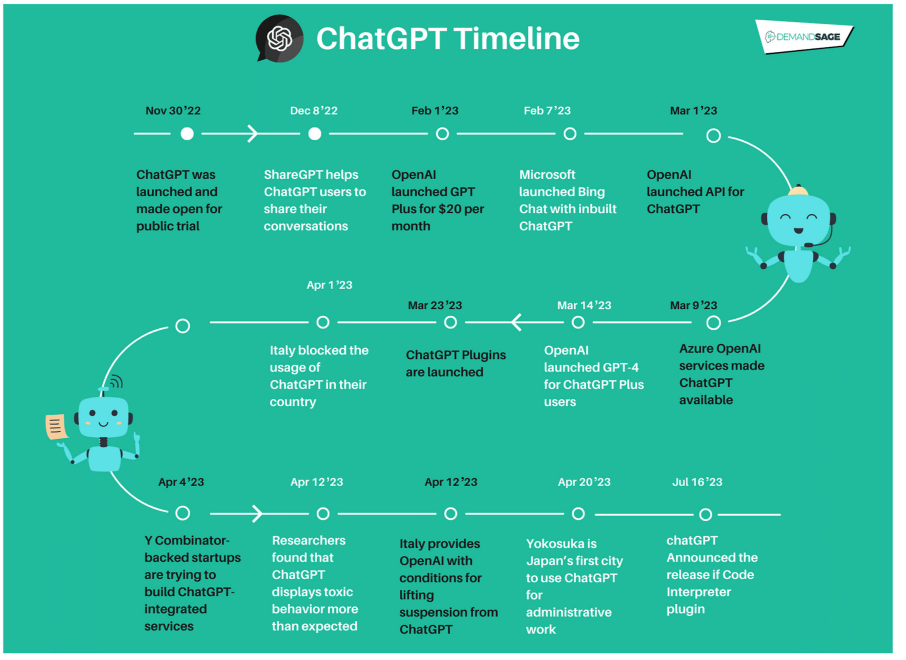


Figure 2.
Overview of ChatGPT statistics it is time to explore all the figures around the AI Chatbot in a detailed manner

Source(s): Demandsage chatgpt-statistics (2023)

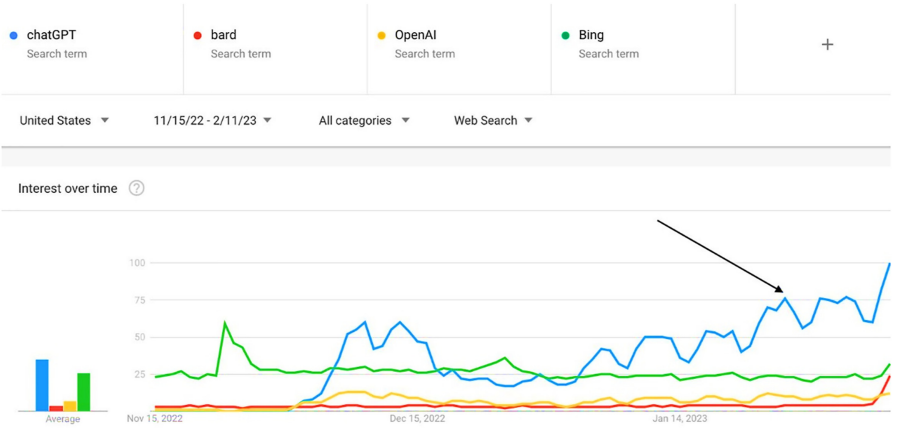
Cooper (2021), a zero-shot setting can be considered challenging in an unfair way because even humans, including our students, benefit from having examples before attempting a task.

The development of FS learning is greatly boosted by the significant increase in the model size of PT-3 (Brown *et al.*, 2020). The upcoming GPT-4, anticipated to be launched in 2023 alongside ChatGPT-4, is set to feature around 100tn parameters. This remarkable development is noteworthy as it is around 500 times more parameters than in GPT-3, bringing it closer to the neural connection count in the human brain. This information has been cited in various sources such as Kovanovic (2022). There is speculation that if the scale is increased by another 100-fold (similar to the variance between GPT-2 and GPT-3), FS performance could potentially approach human-level capabilities. Lauret (2020) postulated an supposition, which may not be accurate, that human neurons correspond roughly to neural network parameters. From this perspective, Lauret (2020) recommended that the human brain contain about 100 billion neurons and forms approximately 100–500tn synaptic connections. Cooper (2021) proposed that if scaling is the key to achieving human-like intelligence, then GPT-3 falls short by a factor of 1,000. It is significant to note that language models like GPT-3 and its contestants are not designed to store or retrieve factual information and lack conventional knowledge. Their strength lies in predicting the next words in a sequence.

ChatGPT

OpenAI unveiled its latest AI chatbot, ChatGPT, through a free preview on November 30, 2022. Hao (2022), OpenAI, 2022 and Jin and Kruppa (2023), this move were significant as it resulted in OpenAI's estimated value surging to US$29bn. A chatbot is a software application that uses AI to simulate human-like conversations. A chatbot is a type of software that utilizes AI to converse with people in a way that is similar to how humans communicate. It can promptly provide responses to users' inquiries or requests as reported by (Murati in 2022). ChatGPT managed to attract a million users within just five days of being released. Figures 3–5 presents the rating interest of Al, ChatGPT, Bard and Bing, respectively. Proper credit line has been presented.

ChatGPT is developed using OpenAI's GPT-3 series of wide language models and is enhanced through a combination of guided and reinforcement learning techniques.

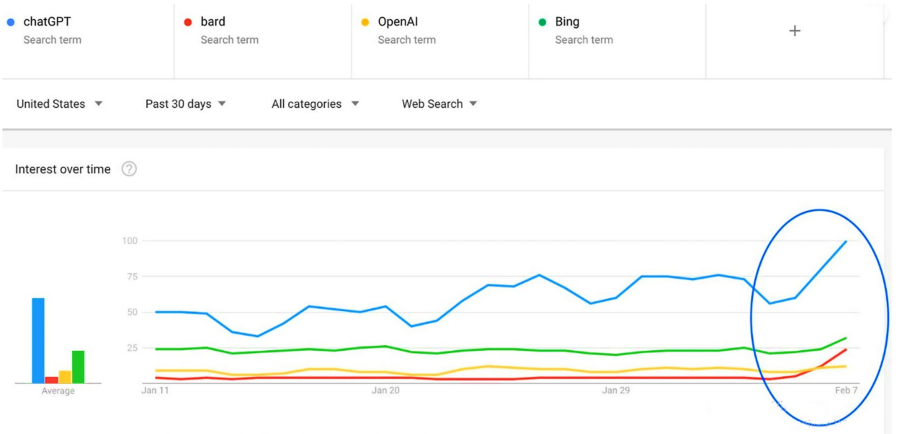


Source(s): Google Trends for the US. (2023)

Figure 3.
Looking closer, interest in OpenAI (the company behind chatGPT) (the yellow line on the graph above) remains stable (albeit at an elevated level vs before)

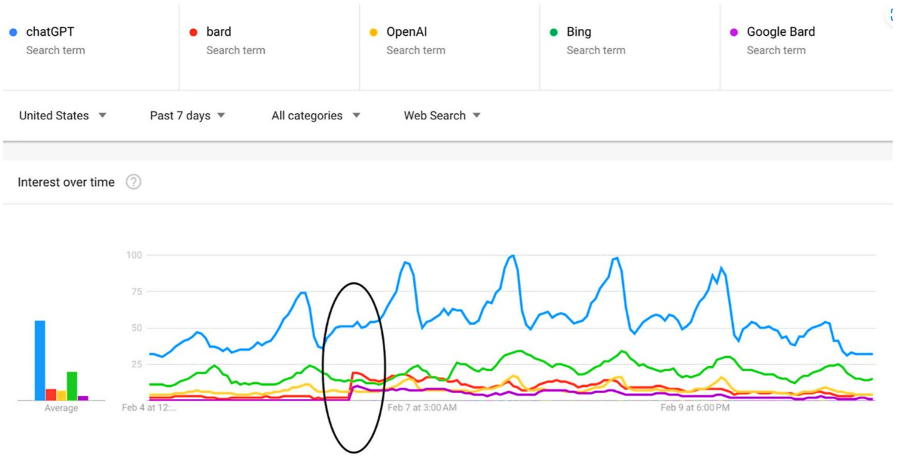
In contrast to search engines like Bing, Google and Baidu, ChatGPT doesn't scour the Internet for up-to-date details, and its empathetic is limited to pre-2021 knowledge. This limitation has led to noticeable inconsistencies in its factual correctness, which was highlighted as a notable disadvantage in a study (Vincent, 2022).

ChatGPT, an OpenAI creation belonging to the GPT-3 lineage, is honed through supervised and strengthening learning techniques. Unlike Internet search engines such as Google, Bing or Baidu, ChatGPT doesn't scour the web for the newest updates. Instead, its scope of understanding is confined to information amassed prior to 2021. Consequently, its accuracy in terms of facts can fluctuate, a drawback acknowledged by Vincent in 2022. The opening section highlights that ChatGPT gained significant popularity due to its comprehensive replies and eloquent solutions across various fields of expertise. It was acknowledged that it was the initial instance when such a potent Chatbot was accessible to everyone through a straightforward and costless web interface (Rose, 2022).



Source(s): Google Trends for the US. (2023)

Figure 4.
Surprisingly, while Microsoft had a good event on Feb 7, the interest in Bing Search didn't explode over the past few days. I don't know why, probably because access to the new Bing Chat is not yet available to the public in mass. So beyond key reporters, most people don't have any experience with Bing Chat



Source(s): Google Trends for the US. (2023)

Figure 5.
Zooming into the past seven days, this is what you see from Feb 04 until now. There was a big (relative to before) bump for Google bard and a generally increasing interest in Bing. Interest in ChatGPT doubled in the US over the past two weeks (Late Jan To Early Feb 2023) AI tool tracker

What opportunities does ChatGPT offer for higher education?

AI language models offer a significant benefit in terms of asynchronous communication, which has been observed to enhance student cooperation and engagement. The capability of this model enables students to interact by posting queries and discussing topics without the need to be present simultaneously (Li and Xing in 2021). ChatAPIs offer several assistance, one of which is promoting collaboration between students. For instance, chat APIs can be used to establish student groups, thereby enabling them to work together on assignments and projects. Additionally, chatAPIs can facilitate remote learning, which is particularly helpful for students who cannot attend classes due to mental or physical health concerns (Barber *et al.*, 2021; Lewis, 2022).

Evaluation is a crucial aspect of higher education that enables the assessment of student progress and learning. A variety of assessment methods are employed, such as exams paper, projects and demonstrations, to measure different learning outcomes, including knowledge, skills, behavior and attitudes. GPT-3 has the potential to be used in higher education to create personalized assessments. By using GPT-3, tailored exams or quizzes can be generated for each student, considering their specific needs and abilities (Zawacki-Richter *et al.*, 2019; Barber *et al.*, 2021). According to Bommasani *et al.* (2021), GPT-3 has the potential to be extremely helpful in language or critical thinking courses. By generating customized questions, GPT-3 could encourage students to showcase their abilities and knowledge. GPT-3 has the potential to utilize in higher education for the development of communicating assessments in the form of games. This could involve the creation of chatbots or virtual assistants that use natural language interaction to present students with questions or problems to solve.

What difficulties does the assessment of GPT-3 pose in the context of higher education?

ChatGPT has the proficiency to provide numerous advantages for evaluating students in higher education, there are several significant footraces that must be considered. One issue with utilizing AI language models, such as ChatGPT and others like it, for higher education assessment, is the potential for academic dishonesty, specifically plagiarism. The objective of AI essay writing software is to create essays by following specific guidelines or prompts. As a result, students may use such software to submit essays that are not their own work, which is considered cheating and reverses the primary aim of higher education to challenge and teach students (Dehouche, 2021). This could result in degrees losing their value.

GPT-3 postures a difficulty in that it may be utilized to give special treatment to certain students. If a student can employ GPT-3 to produce excellent written work, they may gain an unjust advantage over their peers who cannot access the tool. Subsequently, this could lead to imbalances in the evaluation procedure. Assessing a student's comprehension can be challenging when they use a chatbot application to provide answers to questions. Cotton *et al.* (2023), academic staff may struggle to differentiate between the student's original work and the chatbot's responses. The reason for this difficulty is that the answers given by the chatbot may not accurately reflect the student's true level of understanding.

In what ways could the faculty of a university create evaluations that would impede or decrease the reliance of students on ChatGPT?

University staff can employ several effective strategies to design assessments that discourage or limit the use of ChatGPT by students. One such strategy is to devise assessments that necessitate students to exhibit their critical thinking, problem-solving and communication abilities (Greenstein, 2012). He further stated that instead of simply

asking students to write an essay on a given subject, university staff could create assessments that mandate students to participate in group discussions, presentations or other interactive activities that involve the practical application of their knowledge and skills. By doing so, it can become more challenging for students to rely on other AI language models or ChatGPT to complete their tasks and it can also foster critical thinking and self-directed learning.

Academic writing is expected to acknowledge the work of others with accurate citations and references, which helps to credit the original authors and validate the research (Salvagno *et al.*, 2023). Conversely, AI language models like ChatGPT may not provide proper referencing, as they may not have access to relevant sources or the correct citation formatting. To avoid students from relying solely on AI language models, educators can design open-ended assessments that require originality and creativity, such as developing and defending their own arguments or research questions. This tactic makes it harder for students to use AI language models to complete their assignments as it necessitates critical thinking and original ideas. Another way is to incorporate in-class discussions, group work or presentations that require real-time participation and collaboration. This reduces the likelihood of using external AI tools for the entire assignment (Kohnke and Moorhouse, 2021). Finally, to ensure that students do not use AI language models during assessments, real-time or proctored exams can be used.

In what way could academics utilize ChatGPT to avoid students committing plagiarism?

Teachers can utilize ChatGPT to avoid students committing plagiarism by using it as a tool to check the originality of their work. ChatGPT can analyze students' writing and compare it to a large database of existing work to identify any potential matches or resemblances. This can help educators to quickly address and identify any cases of plagiarism, and encourage students to produce original work. Academics can utilize ChatGPT to avoid students committing plagiarism by using it as a tool to perceive potential instances of plagiarism in student assignments. By inputting student work into ChatGPT, the model can analyze the text and compare it to a vast database of existing content to determine whether any portions of the work have been copied. This can help educators identify instances of plagiarism and provide appropriate feedback to students to encourage original writing. ChatGPT can be utilized by academics to help students avoid committing plagiarism in other several ways:

ChatGPT can help students to paraphrase their work by suggesting alternative ways of phrasing a sentence or paragraph. This can help them to avoid inadvertently using the same words or sentence structure as the original text. ChatGPT can be used as a plagiarism detection tool to check whether a student's work contains any text that is identical or similar to existing online sources or other academic publications. This can help to identify potential cases of plagiarism before they are submitted. ChatGPT can provide students with guidance on how to conduct thorough research, cite sources and avoid plagiarism. This can include providing tips on how to use different search engines, databases and academic journals. ChatGPT can provide students with guidance on how to properly cite sources in their work, including different citation styles like American Psychological Association (APA), Modern Language Association (MLA) or Chicago. This can help students to avoid plagiarism by ensuring that they give proper credit to the original author. Teachers can conduct oral assessments, presentations, or interviews where students have to explain their thought processes and defend their ideas. This makes it difficult to rely solely on AI-generated text Chen *et al.*, 2022. Another way is to emphasize critical thinking and analytical skills in the curriculum. Encourage students to analyze, synthesize, and apply knowledge rather than relying on simple content generation (Spector and Ma, 2019).

Future scope

The ChatGPT language model has enormous potential for future growth and expansion. Some possible areas of development include enhancing its natural language understanding to better understand contextual clues and nuances in language, resulting in more precise and relevant responses. Personalization of the conversational experience by training ChatGPT on individual user preferences and past interactions is another option. Incorporation with various smart devices would enable seamless communication with connected devices in offices, homes and elsewhere. Developing ChatGPT to understand and respond in multiple languages would make it a valuable tool for global communication. Additional training in specific domains such as medicine, law or finance would enable ChatGPT to provide accurate and useful information to users in these fields. ChatGPT can also be improved to recognize and respond to emotional cues in language, making it more empathetic and human-like in its interactions. Collaboration with other AI systems would provide more comprehensive solutions to complex problem all overall, ChatGPT has the potential to transform human-machine interactions and communication and we can expect further growth and innovation in this area in the future.

Conclusion

AI is becoming more mainstream with the emergence of technologies such as GPT-3 and ChatGPT (Vincent, 2022). The full impact of this tendency on society as a whole, and on higher education specifically, is yet to be fully understood. We have just started to witness the impact of this technology on the world as a whole, and specifically on higher education. The potential realization consciousness of true AGI within the next ten years, as suggested by Ortiz (2022), would result in significant societal consequences. The potential advancements in the future, it is inconsistent that AI-powered anti-plagiarism tools are used to evaluate the originality of academic work, while other AI technologies, such as ChatGPT, can quickly circumvent plagiarism detection software. The irony deepens when considering that GPT-3 can even write a review of a student's AI-generated assignment on behalf of a teacher with a simple command (Sharples, 2022). These results in one AI circumventing another AI, with humans simply pressing a few buttons and no actual learning taking place.

To sum up, higher education can benefit from using ChatAPIs and GPT-3 by improving student collaboration, engagement and accessibility. ChatAPIs can support asynchronous communication, enable student groups, provide timely feedback and facilitate remote learning. GPT-3, on the another side, can be used for various applications such as, text summarization, language translation, question answering, personalized assessments and text generation,. Despite the potential advantages, there are also concerns related to academic dishonesty and plagiarism as these tools may be used to enable cheating, and it can be challenging to differentiate between machine-generated writing and human. Therefore, universities need to carefully consider the potential risks and rewards of using these tools and establish ethical and responsible policies and procedures for their use. They may also provide training and support for students and faculty and use multiple methods to spot and prevent academic dishonesty. By doing so, universities can make the most of ChatAPIs and GPT-3 while preserving the integrity of their assessments and the quality of their educational programs.

Recommendation for higher education teachers, students and institutions

These are our early ideas regarding suggested actions for faculty members, students and institutions. This paper discusses various solutions to the problem of students using AI text

generators such as ChatGPT and GPT-3 to write their academic assignments. One solution is to use physical closed-book exams or online proctoring/surveillance software. However, this approach has been criticized for indorsing memorization of less-than-useful information rather than evolving employable skills (Van *et al.*, 2016). An alternative approach involves creating writing tasks that challenge AI systems in areas where they currently lack proficiency, such as evaluating images and videos, analyzing class discussions, and discussing contemporary events not covered in the text generator's training data. Mills (2023) proposes evaluating students' capacity to create various sources and articulate unique lookouts. Currently, ChatGPT lacks the ability to include in-text citations and a bibliography in its output, representing a significant drawback for academic writing tasks. Nevertheless, OpenAI has developed a prototype called WebGPT, which can utilize web browsing to validate sources and quotes. In the interim, Elicit is a GPT-3-based tool that can suggest academic articles and provide summaries from a repository of 175m scholarly papers (Andrews, 2023).

In this study we presented some recommendations for students who are often more adept at using technology than their teachers. The recommendations include being aware of academic integrity policies, becoming digitally literate and mastering AI tools, using AI as a tool to improve writing skills and generate new ideas, being cautious of unreliable sources and misinformation, improving critical and creative thinking by reading widely. We deliberate the importance of incorporating AI tools in digital literacy education, including Elicit and Grammarly, and provide faculty training on tools such as ChatGPT. It also highlights the need to provide training on academic integrity to develop educational modules or workshops on academic integrity and proper research techniques. Help students understand the value of authentic learning and the consequences of plagiarism (Zeb *et al.*, 2023; Mahabeer and Pirtheepal, 2019). Lastly, this paper emphasizes the need for further research on the effects of AI tools on learning and teaching.

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